The Scientific Creative Thinking Test (SCTT): Reliability, Validity, and Automated Scoring

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Abstract

Creative thinking is a primary driver of innovation in science, technology, engineering, and math (STEM), allowing students and practitioners to generate novel hypotheses, flexibly connect information from diverse sources, and solve ill-defined problems. To foster creativity in STEM education, there is a crucial need for assessment tools for measuring STEM creativity that educators and researchers can apply to test how different teaching approaches impact scientific creativity in undergraduate education. In this work, we introduce the Scientific Creative Thinking Test (SCTT). The SCTT includes three subtests that assess cognitive skills important for STEM creativity: generating hypotheses, research questions, and experimental designs. In five studies with young adults, we demonstrate the reliability and validity of the SCTT—including test-retest reliability and convergent validity with measures of creativity and academic achievement—as well as measurement invariance across race/ethnicity and gender. In addition, we present a method for automatically scoring SCTT responses, training the large language model Llama 2 to produce originality scores that closely align with human ratings—demonstrating STEM-specific, automated creativity assessment for the first time. The full SCTT, along with the code to automatically score it, are available on a repository in the Open Science Framework.

Keywords: creativity assessment; creative thinking; large language models; scientific creativity; STEM education

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Creative thinking is central to academic and career success in Science, Technology, Engineering, and Math (STEM). Creativity is consistently among the top 21st century skills, and cultivating creativity is a primary goal of modern STEM education (Turiman et al., 2012; Yang, 2022). There is thus a strong demand for tools to measure STEM creativity that allow researchers and educators to determine what works (and what doesn't work) in STEM education to promote creative thinking. Moreover, such tools need to be scalable and efficient, as teachers often lack the time and resources to manually score hundreds or thousands of creative responses. Here, we introduce the Scientific Creative Thinking Test (SCTT), an assessment designed to measure core cognitive competencies of scientific creativity (e.g., hypothesis generation) for STEM undergraduate education. We conduct five studies with undergraduate students to assess the reliability and validity of the SCTT, and test a new method to automatically score SCTT responses using large language models.

Scientific Creative Thinking

Creativity[1] has been studied far more extensively in the arts than the sciences (Saptono & Hidayah, 2020), potentially due to the perception that science requires technical rather than creative skills (Loehle, 1990). Yet creativity plays a pivotal role in scientific pursuits, requiring the generation of original hypotheses, the integration of diverse sources of information, and the ability to solve ill-defined problems (Daker et al., 2022; DeHaan, 2011). Moreover, although the domain-specificity of creativity is debated, scientific innovation clearly requires domain knowledge to generate ideas that are both novel and scientifically plausible. Thus, developing measurements tailored to evaluating creativity in science is key to understanding how creative potential manifests in STEM and guiding education to promote it.

[1] We use the term "creativity" throughout the manuscript to refer to divergent thinking: the production of original and effective ideas. This operationalization of creativity is thus specific to early-stage divergent thinking and does not account for later stages of the creative process.

Current psychometric tests of scientific creativity include the Scientific Creativity Test (Hu & Adey, 2002), the Divergent Problem Solving Ability in Science test (Aschauer et al., 2022), and the Creativity Scientific Ability Test (Sak & Ayas, 2019). Each test was designed to measure different aspects of scientific creativity with students from different educational levels. All three tests are based on the concept of divergent thinking—the ability to generate multiple possible solutions to a problem or question (e.g., Guilford, 1967). However, they also differ in some key aspects, including the theoretical framework for conceptualizing scientific creativity.

Hu and Adey (2002) characterized scientific creativity as the ability to generate an original product of social or personal value. They proposed a Scientific Structure Creativity Model that incorporates three facets: scientific process (scientific thinking and imagination), product (technical products, scientific knowledge), and traits (fluency, flexibility, and originality of ideation). Using this framework, the authors developed the Scientific Creativity Test (SCT) for secondary school students. The SCT is similar to classic tests of divergent thinking, such as the Torrance Test of Creative Thinking (e.g., the product improvement task; Torrance, 1966), but within scientific contexts. In terms of psychometric properties, the SCT was initially validated via face validity of STEM teachers, and by comparing performance of students from different age groups. Other work has since reported correlations between SCT performance and personality traits linked to creativity in high school students, including creative self-efficacy (Qiang et al., 2020).

Scientific creativity has also been cast as a dual process involving generating hypotheses and conducting experiments (Klahr & Dunbar, 1988). Using this framework, Sak and Ayas (2013) devised the Creative Scientific Ability Test (C-SAT), which consists of three science-related abilities: hypothesis generation, experimental design, and evidence evaluation. The C-SAT has a domain-specific focus, covering STEM topics such as biology, chemistry, physics, and ecology. Sak and Ayas (2013) assessed the reliability and validity of the C-SAT in a sample of middle school students, reporting internal consistency with a single factor

comprising all three subtests (hypothesis generation, experiment design, and evidence evaluation).

More recently, Aschauer et al. (2022) devised the Divergent Problem Solving Ability in Science (DPAS) test. The DPAS consists of two subtests: the Divergent Ideation in Science Tasks (DIST) and the Divergent Ideation in Experimental Tasks (DIET). The DIST subtest is similar to the SCT by Hu and Adey (2002) and other divergent thinking tests, e.g., generating reasons for phenomena like a non-freezing ice cream. The DIET subtest focuses on experimental skills and domain specific knowledge, e.g., determining the height of 150cm without a measuring tape. In a large sample of secondary students, Aschauer et al. (2022) demonstrated the construct validity and reliability of the DPAS scores, as well as factorial longitudinal invariance and group invariance in a classroom intervention study: students receiving a scientific creativity curriculum, Flex-based Learning (Haim & Aschauer, 2022), increased their DPAS scores more than a control group. Notably, the DPAS is scored for fluency and flexibility, not originality, consistent with the goals of the Flex-based Learning curriculum to boost the number and variety of ideas.

Domain-specific assessments have also been developed for specific STEM domains, such as psychological science (Sternberg et al., 2020). Sternberg and colleagues (2017) administered a scientific creativity test to a sample of college students, consisting of three open-ended questions. The questions asked students to formulate their own research question relating to human behavior/psychology, propose hypotheses to explain the phenomenon, and design an experiment to test their hypotheses. Responses were scored holistically on a scale from 1 (unsatisfactory) to 5 (outstanding) based on the quality of the responses. These test scores correlated with measures of scientific reasoning, such as drawing scientific conclusions (Sternberg et al., 2019), demonstrating convergent validity. However, the test focused solely on psychology and used only a single item, reducing reliability and generalizability.

Automated Creativity Scoring

Scoring creative thinking tasks presents a major challenge for educators and researchers in terms of human resources. Many divergent thinking tasks, including the scientific creativity tasks mentioned above, require manually counting and coding hundreds or thousands of responses (Kleinkorres et al., 2021)—an untenable commitment for many under-resourced educators and researchers without access to trained raters who can help with scoring. To address the challenges of manual scoring, researchers are developing computer-based methods that can automatically score creative thinking tasks using machine learning and artificial intelligence (AI; Beaty & Johnson, 2021; Buczak et al., 2023; Dumas et al., 2021; Johnson et al., 2022).

Recently, Organisciak et al. (2022) trained an AI system—the Generative Pretrained Transformer (GPT), a large language model capable of achieving human-level performance on a range of tasks—to automatically score the originality of responses to the alternate uses task (e.g., Guilford, 1967), showing robust prediction of human ratings that far exceeded the state-of-the-art, computer-based scoring method (i.e., semantic distance). GPT was "fine-tuned" to perform the task of rating originality—seeing thousands of AUT responses and human ratings, allowing it to learn how humans evaluate originality on this task. Other work has shown similarly strong performance of language models to score verbal creativity tasks, including metaphor production (DiStefano et al., 2024) and creative problem solving (Luchini et al., 2023). Despite the potential of language models to automate creativity assessment, no such methods currently exist to automatically score domain-specific creative thinking tasks such as scientific creativity.

The Present Research

To foster creativity in STEM education, it is essential to have assessment tools that measure creative thinking reliably and validly. Researchers have developed tests of scientific creative thinking, with many demonstrating psychometric strengths. However, current tests

have largely focused on K-12 samples, with very little work on undergraduate students—a population at the cusp of entering professional or academic research fields. This gap is particularly concerning given the unique challenges and opportunities presented at this educational stage, where students transition from structured learning to more independent, creative thinking and problem solving required for scientific inquiry. Moreover, current scientific creativity tests rely on labor-intensive, manual scoring, which places a substantial burden on teachers and researchers, hindering scalability and practical utility.

In the present research, we introduce the Scientific Creative Thinking Test (SCTT). The SCTT was designed to evaluate three skills critical in the natural sciences: the formulation of research questions, the generation of hypotheses, and the design of experiments. We aimed to produce items that adhere to a consistent response format: participants are presented with an open-ended problem and instructed to produce novel and scientifically plausible solutions.

Our goal was to assess creative potential in STEM while requiring minimal expertise within a particular STEM field. Having a rigorously validated test of scientific creativity will allow educators to test whether specific teaching strategies are successful in fostering creative thinking skills relevant for science (hypothesis generation, question asking, experiment design)—independent of their current level of course content knowledge. The SCTT is thus intended to measure general scientific creative thinking; it does not require domain-specific knowledge of any particular field (e.g., chemistry, biology) but rather measures the skill of scientific creative thinking.

We conducted five studies with undergraduate students to assess the reliability and validity of the SCTT. Study 1 focused on test construction and reliability; Study 2 tested criterion validity; Study 3 assessed test-retest reliability; and Study 4 examined measurement invariance across genders and race/ethnicity. In Study 5, we introduce an automated scoring system for the SCTT to enhance scalability and practical utility. Building on recent work leveraging large language models to score creativity tasks (e.g., DiStefano et al., 2024; Luchini et al., 2023;

Organisciak et al., 2023), we fine-tuned an open-source language model, Llama-2, to generate SCTT creativity scores that align with human ratings—aiming to provide the first STEM-specific method for automated creativity scoring.

Study 1

The goal of Study 1 was to generate and psychometrically evaluate items for the SCTT. We generated items for the three SCTT subtests—research questions, hypotheses, and experiments—and administered them to a sample of undergraduate students. Responses were rated for creative quality by independent raters using a subjective scoring method (Hass et al., 2018; Silvia et al., 2008). A confirmatory factor analysis was specified to assess how well each item loaded onto its respective factor (research questions, hypotheses, and experiments).

Method

Participants

A total of 151 undergraduate students from Pennsylvania State University (PSU) completed the study (mean age = 18.47, SD = 1.26; 122 females, 17 males, 0 other gender, 12 preferred not to disclose)[2]. Students received course credit for their participation, and the study was approved by the PSU Institutional Review Board.

SCTT Construction

First, a list of potential SCTT items was produced by graduate students, advanced undergraduates, and faculty (including co-authors of this manuscript) for all three SCTT categories. Then, the first and senior authors refined the item list, selecting a reduced set of items for testing. Our goal was to include four items per SCTT subtest; this would allow us to construct two alternate test forms (in Study 3), with two items per form, thereby facilitating longitudinal and intervention studies. We included a fifth item per SCTT subtest to allow one

[2] Due to an issue with the experiment software, other demographic information for this sample was not collected. However, since the sample was collected from the same population of undergraduate students as Studies 2 and 3, similar characteristics can likely be assumed.

item to be excluded in case it did not meet psychometric criteria. The 15 candidate items are listed in the Appendix. The task was administered online via Pavlovia (https://pavlovia.org).

Research question generation is evaluated by providing an environment and asking participants to formulate potential research questions to learn about the environment. For example, “You have just discovered a jungle with no human life that the scientific community had no idea existed. What scientific questions could you ask to learn about this new jungle?” Example responses include “how do humans affect the food chain in the jungle?” and “does the average yearly rainfall in the area have correlation to the nesting habits of birds in the jungle?”.

Experimental design is evaluated by presenting a phenomenon along with a hypothesis about the cause or consequence of the phenomenon. The participant is then tasked with devising an experiment to test the given hypothesis. For example, “You think birds can understand humans but just choose to ignore us because what humans say isn’t important to their lives. How could you test that hypothesis?” Example responses include “hide some food and tell the bird where you hid it” and “put a bird in a cage and explain a task that it must complete to be set free.”

Hypothesis generation is evaluated by asking participants to formulate causes or potential effects in response to a given situation, event, or phenomenon. For example, “You notice that the water in one lake is warmer than the water in another lake even though they both get the same amount of sunlight. What hypotheses do you have about why that is?” Example responses include “there are geothermal vents in the warmer lake” and “the altitude for the two lakes may be much different, causing a temperature disparity.”

For each item, participants produced three responses (Barbot, 2018). The number of responses was fixed to control for fluency confounds (Forthmann, Szardenings, et al., 2020; Silvia et al., 2013) and because the test was designed to assess originality rather than fluency. Task and item order were randomized for each participant.

Subjective Scoring

SCTT responses were scored for creative quality using a subjective scoring method (Hass et al., 2018; Silvia et al., 2008). Six undergraduate students were trained to serve as raters. They were instructed to consider both the originality and plausibility of the responses, using a 1 (*not at all creative*) to 5 (*very creative*) scale, based on an adaptation of the Silvia et al. (2008) guidelines. Given the large volume of responses, we used a planned missing design to reduce the rater burden (Forthmann et al., 2023; Graham et al., 2006).

Analysis Plan

Our main goal was to identify reliable items for inclusion in the SCTT. To this end, we used confirmatory factor analysis (CFA) to test how well each item loaded onto its respective factor: hypotheses, questions, and experiments. We assessed model fit for a correlated factors model (3 SCTT latent variables) vs. a higher order factor model (a single SCTT variable indicated by the 3 lower order variables). Specifically, we assessed the model's χ^2 statistic and fit indices RMSEA (should be < .06 to indicate good model fit to the data), SRMR (should be < .08), and CFI (should be > .95) using established cut-off values (cf. West et al., 2012
). We expected the three factors to be moderately to highly correlated, and thus to load highly onto a higher-order SCTT factor.

Results

Descriptive statistics for the SCTT items are reported in Table 1. Overall, participants produced 7,923 SCTT responses to the 15 SCTT items. The intra-class correlation (ICC) between

Table 1

Descriptive statistics for the SCTT items

Variable	M	SD	Min	Max
ed1_birds	2.41	0.63	1.00	3.83
ed2_flower	2.54	0.62	1.00	4.00
ed3_friendly	2.55	0.57	1.00	4.17
ed4_humor	2.48	0.70	1.00	4.33

ed5_mind	2.03	0.63	1.00	4.17
hg1_dogs	2.71	0.71	1.00	5.00
hg2_hole	2.62	0.66	1.00	4.67
hg3_lake	2.73	0.68	1.00	4.83
hg4_sing	2.65	0.58	1.17	5.00
hg5_star	2.42	0.78	1.00	5.00
rq1_island	2.52	0.66	1.33	4.75
rq2_jungle	2.35	0.56	1.52	4.00
rq3_ocean	2.33	0.56	1.44	3.89
rq4_planet	2.35	0.45	1.56	3.78
rq5_robot	2.65	0.47	1.67	3.94

Note. ed = experiment design; hg = hypothesis generation; rq = research question. *N* = 121.

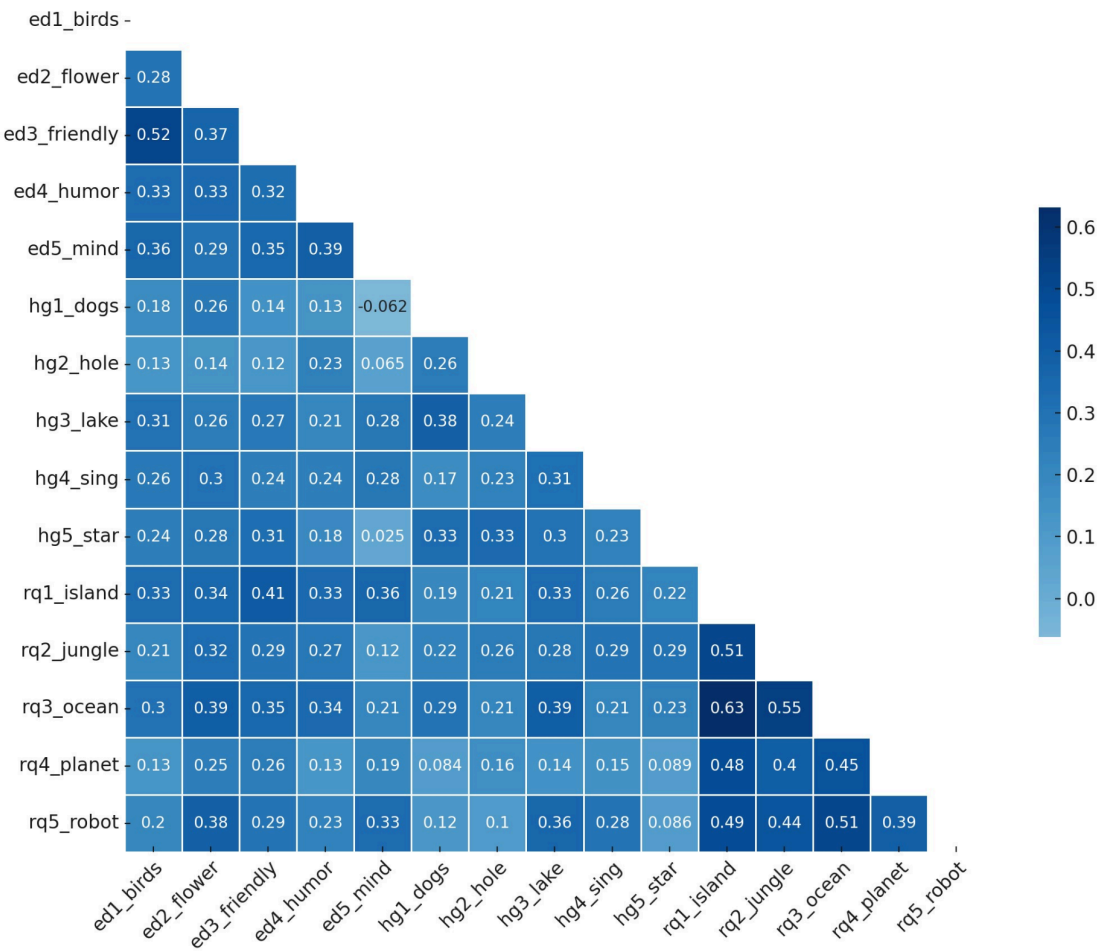
6 raters was computed to evaluate rater agreement across all items and subtasks. Missing values were estimated for incomplete datasets. Good reliability was observed (ICC = 0.74, 95% CI: [0.74, 0.76]) for the absolute agreement on the average ratings using a two-way model. Pearson correlations between the SCTT items are shown in Figure 1.

Next, we defined a measurement model encompassing the three SCTT variables: hypotheses, questions, and experiments. This model fit the data well: $\chi^2(87)$ 102.99, p = .12; CFI .97; RMSEA = .04, 90% CI: [.00, .06]; SRMR = .06. All five indicators showed moderate to high loadings on their corresponding latent variables. Omega reliabilities were acceptable to good (ed: ω = .73; hd: ω = .66; rq: ω = .82). Moreover, large latent correlations were found among the latent variables (questions, hypotheses, and experiments), ranging between .61 to .68. Figure 2 depicts the measurement model and the correlations among the latent variables.

Then, we constructed a higher-order factor model to assess the loadings of the three SCTT variables onto a common factor: $\chi^2(87)$ = 103.00, p = .12; CFI = .97; RMSEA = .04, 90% CI: [.00, .06]; SRMR = .06. Consistent with the high intercorrelations among the three factors, we

Figure 1

Correlations between SCTT Items



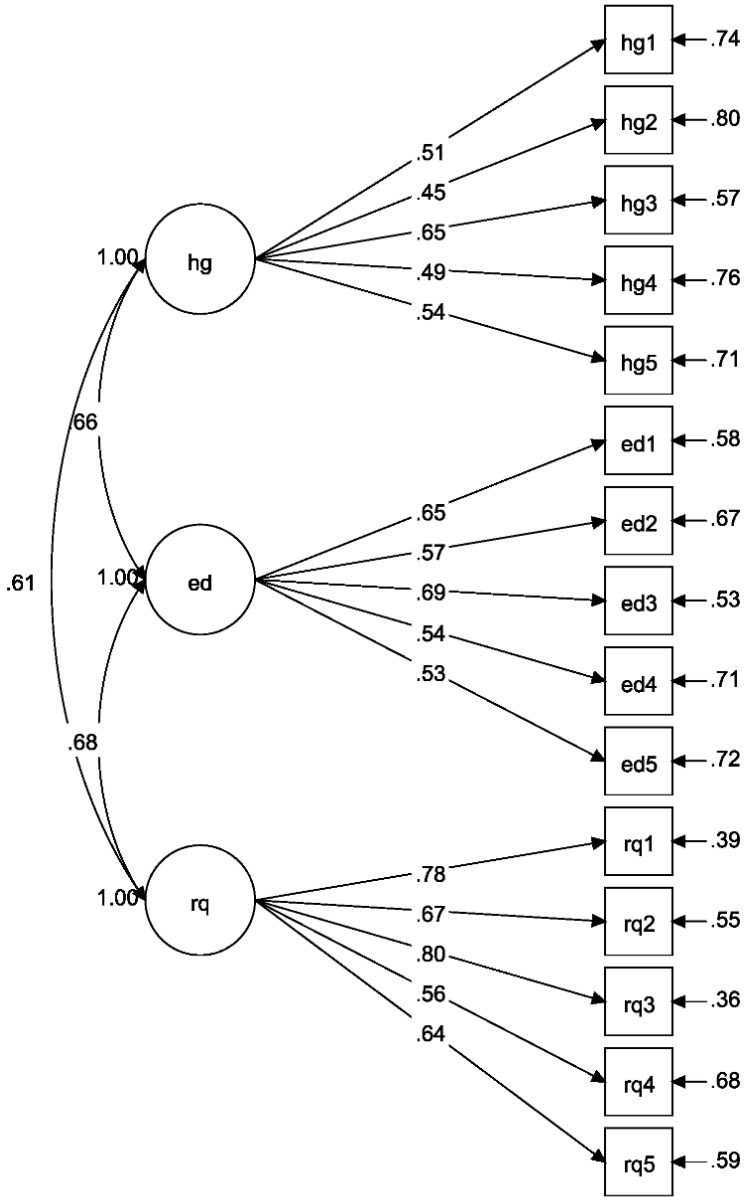
Note. ed = experiment design; hg = hypothesis generation; rq = research question. *N* = 121 (cases with complete data).

found large loadings on the higher-order SCTT factor (see Figure 3). The highest loading was for experiments (.85), followed by hypotheses (.77), and questions (.79). Omega reliability for the higher-order factor was good, ω = .85.[3]

Figure 2

Confirmatory factor analysis for the correlated factors model from Study 1

[3] For the final set of four items for each of the three factors the following omega reliabilities resulted: ed: ω = .70; hd: ω = .63; rq: ω = .81. Omega reliability for the higher-order factor based on the reduced item-set was good, ω = .87. Thus, reliability was highly comparable for the final set of items (cf. Appendix). Complete analyses for the final set of items can be found at the OSF repository.



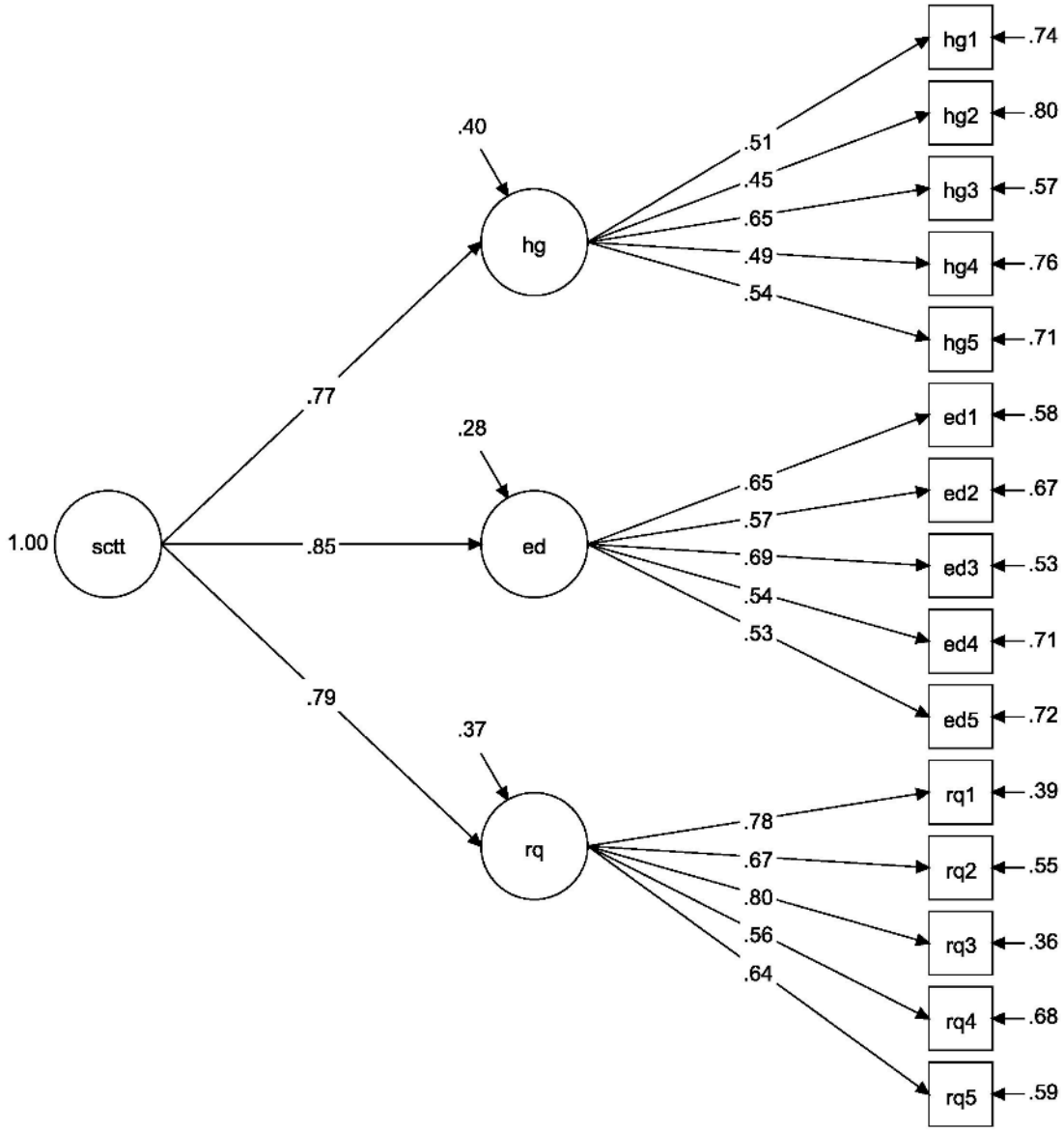
Note. hg = hypothesis generation; ed = experiment design; rq = research question; ed1 = birds, ed2 = flower, ed3 = friendly, ed4 = humor, ed5 = mind; hg1 = dogs, hg2 = hole, hg3 = lake, hg4 = sing, hg5 = star; rq1 = island, rq2 = jungle, rq3 = ocean, rq4 = planet, rq5 = robot.

Study 2

In Study 1, we assessed the reliability of the SCTT—a new test of scientific creative thinking—aiming to assess core competencies relevant for creativity in STEM: generating

Figure 3

Confirmatory factor analysis for the higher-order factor model from Study 1



Note. hg = hypothesis generation; ed = experiment design; rq = research question; ed1 = birds, ed2 = flower, ed3 = friendly, ed4 = humor, ed5 = mind; hg1 = dogs, hg2 = hole, hg3 = lake, hg4 = sing, hg5 = star; rq1 = island, rq2 = jungle, rq3 = ocean, rq4 = planet, rq5 = robot.

research questions, hypotheses, and experiments. We produced five items per competency and found that they all loaded significantly on their respective factors, and when modeled together, the factors showed large loadings on a higher-order SCTT factor.

Having established the reliability of the SCTT, Study 2 aimed to test its criterion validity. To this end, we administered the SCTT to a new sample of undergraduates, who also completed measures of science achievement, general/verbal creativity, and general cognitive ability. We hypothesized that SCTT creativity scores would correlate positively with all three, given prior studies reporting positive associations between creative ability, STEM achievement, and intelligence. Importantly, we further hypothesized that SCTT would have incremental validity by predicting science achievement even after accounting for intelligence, as assessed in a regression model. Finally, we conducted a first exploration of measurement invariance across gender groups.

Method

Participants

The total sample consisted of 212 undergraduates from Penn State University (mean age = 19.22, SD = 3.16; 115 females, 85 males, 3 other gender, 4 preferred not to disclose). Participants received optional course credit for their involvement in the study. The study was approved by the PSU IRB, and participants provided consent prior to completing the study.

Procedure

Students completed the SCTT, as well as a battery of tasks and questionnaires that measure aspects of science achievement (e.g., STEM grades, creative activities and achievements), general/verbal creativity (e.g., divergent thinking, story writing), and general cognitive ability (e.g., fluid and crystalized intelligence). All measures were administered online via Pavlovia (https://pavlovia.org). Additional measures were collected but not analyzed here.

SCTT. Participants completed the three SCTT subtests: research questions, hypotheses, and experiments. For each subtest, we included the four items with the highest factor loadings from Study 1. Participants were randomly assigned to two counterbalanced conditions, which included two SCTT items for each subtest. As in Study 1, for each item, they produced three

responses to control fluency confounds, and responses were scored for creative quality by trained raters.

Science Achievement. Participants completed three measures to assess different aspects of science achievement: science knowledge, creative activities and achievements in science and engineering, and GPA in science courses. To assess science knowledge, students completed two questions from the 2015 PISA Science test: the 'bird migration' question and the 'groundwater extraction' question; responses were scored according to PISA guidelines (Gurría, 2016). To assess creative activities and achievements, they completed the Inventory of Creative Activities and Achievements (ICAA; Diedrich et al., 2018). The ICAA comprises two subscales designed to evaluate both creative hobbies and accomplishments in eight fields encompassing the arts and sciences; we focus on the Science and Engineering subscales for this analysis. Regarding science GPA, students were asked to self-report their grades in science courses at Penn State (Sticca et al., 2017).

Verbal Creativity. We administered three tasks to measure verbal creative thinking: the verb generation task, the AUT, and the five-sentence short story task. The tasks varied in the complexity of responses required—from single words (verb generation) to longer narratives (short stories)—to obtain a comprehensive assessment of verbal creativity. In the verb generation task, participants were presented with nouns and produced creatively associated verbs (20 trials; Prabhakaran et al., 2014); responses were scored for originality using semantic distance via the *SemDis* platform (Beaty & Johnson, 2021). In the AUT, participants generated original uses to 2 objects (box and rope, 2 minutes each); responses were scored for originality using maximum associative distance (MAD), a semantic distance metric that reflects the most distance word (from the cue word; e.g., box) in a multi-word AUT response (Yu et al., 2023). For the short story task, participants were presented with three words (*stamp-letter-send*) and asked to write a creative short story using those words that is about five sentences long (Prabhakaran et al., 2014); stories were scored for originality using divergent semantic

integration (DSI; Johnson et al., 2022), a metric that computes the pairwise semantic distance between all words in a story (using the BERT language model; Devlin et al., 2019), and then averages these distance values to obtain a single DSI score.

General Intelligence. Participants completed a battery of tasks to evaluate three facets of intelligence: fluid intelligence (Gf), crystallized intelligence (Gc), and broad retrieval ability (Gr). Gf was assessed with a series completion task from the Culture Fair Intelligence Test (13 items, 3 minutes), in which participants are presented with a sequence of three changing images (simple line drawings) and asked to choose the subsequent image that follows the rule governing their change (Cattell, 2013). Gc tasks included: 1) the advanced vocabulary test (18 items) and 2) the extended range vocabulary test (Ekstrom, 1976), both of which require selecting a synonym for a given word from a list (8 minutes in total). Gr tasks comprised two category fluency tasks (e.g., Unsworth et al., 2011): listing animals and listing fruits/vegetables (2 minutes each); responses were summed for each task and then averaged to form a composite.

Analysis Plan

Our aim for Study 2 was to assess the criterion validity of the SCTT. We began by specifying a CFA to model the intercorrelations between SCTT performance, science achievement, verbal creativity, and general intelligence. We subsequently ran a regression analysis with SCTT and intelligence as predictors of science achievement scores to determine whether SCTT performance incrementally predicts science achievement beyond intelligence. Model fit was evaluated based on the same criteria as in Study 1.

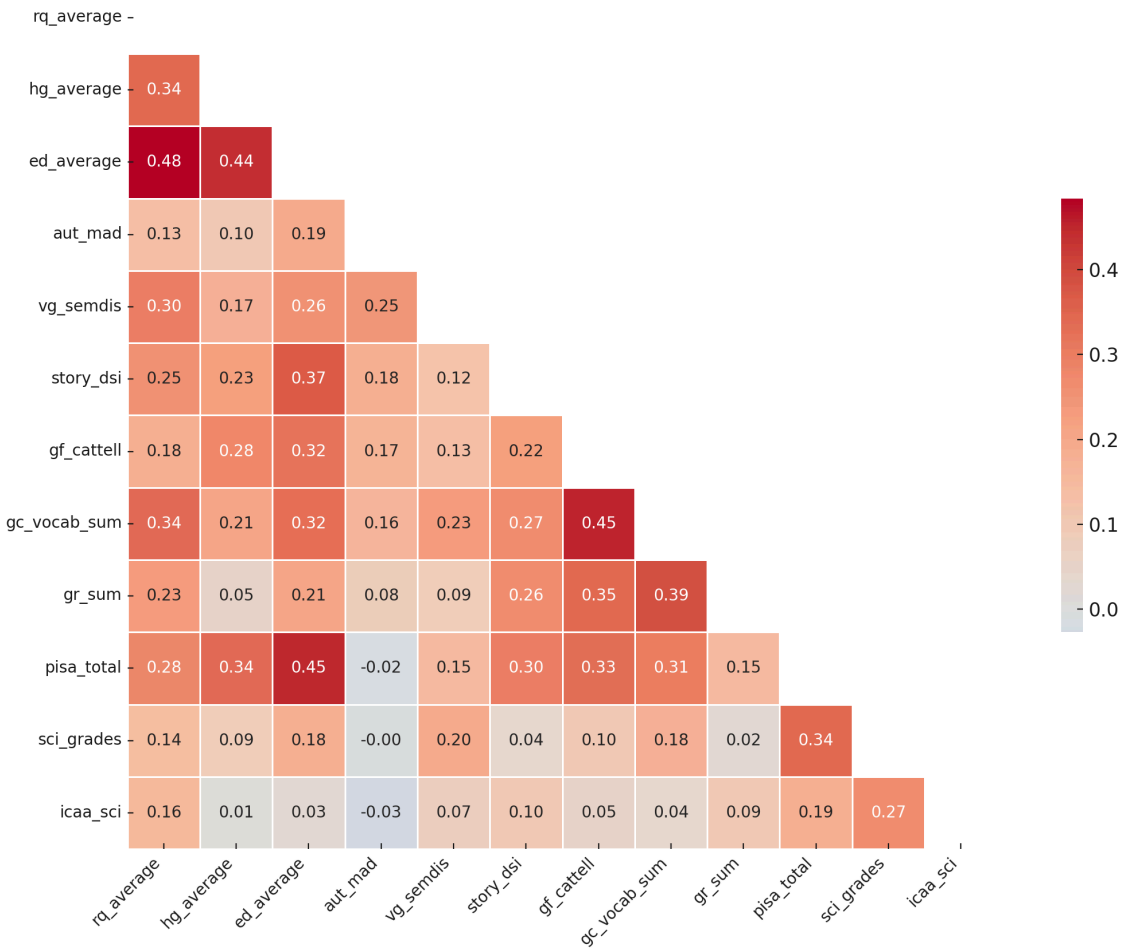
Finally, we evaluated measurement invariance based on Bayesian structural equation modeling. More details for the analytical steps are provided in the OSF in a supplemental material file.

Results

Overall, participants produced 3,291 responses to the 12 SCTT items (6 items per participant). We computed the ICC between the 3 raters to test for rater agreement across all items and subtasks. Moderate reliability was observed, ICC = 0.67, 95% CI: [0.65, 0.69], for absolute agreement using a two-way model. Omega reliability of the SCTT was acceptable with ω = .71. Pearson correlations between the SCTT subtasks and validation measures are shown in Figure 4.

Figure 4

Correlations between SCTT Subtests and Validation Measures

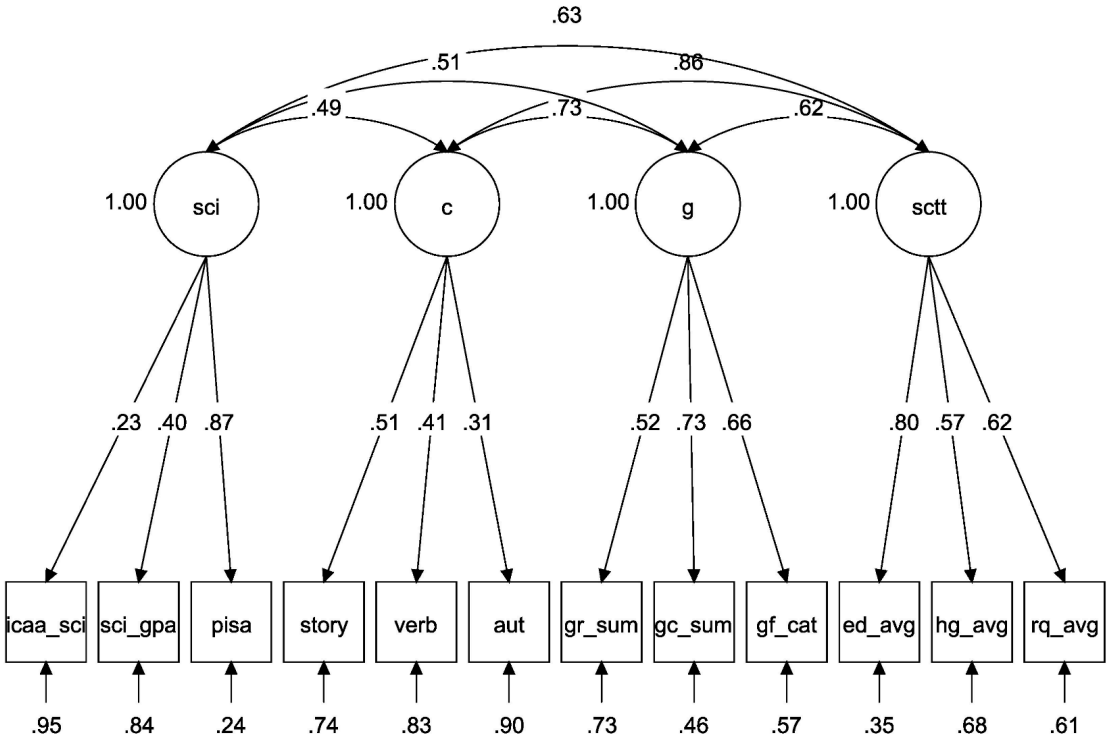


Note. rq_average = research question, average rating; hg_average = hypothesis generation, average rating; ed_average = experiment design, average rating; aut_mad = alternate uses task, maximum associative distance;

vg_semdis = verb generation, semantic distance; story_dsi = story task, divergent semantic integration; gf_cattell = fluid intelligence, Cattell's culture fair test score; gc_vocab_sum = crystallized intelligence, vocabulary advanced and extended range tests, summed; pisa_total = PISA (Programme for International Student Assessment), total score; sci_grades = science grades; icaa_sci = Inventory of Creative Activities and Achievements, science score. N = 163 (sample size with no missing data). *N* = 163 (cases with complete data).

Figure 5

Confirmatory factor analysis of the correlated factors model from Study 2

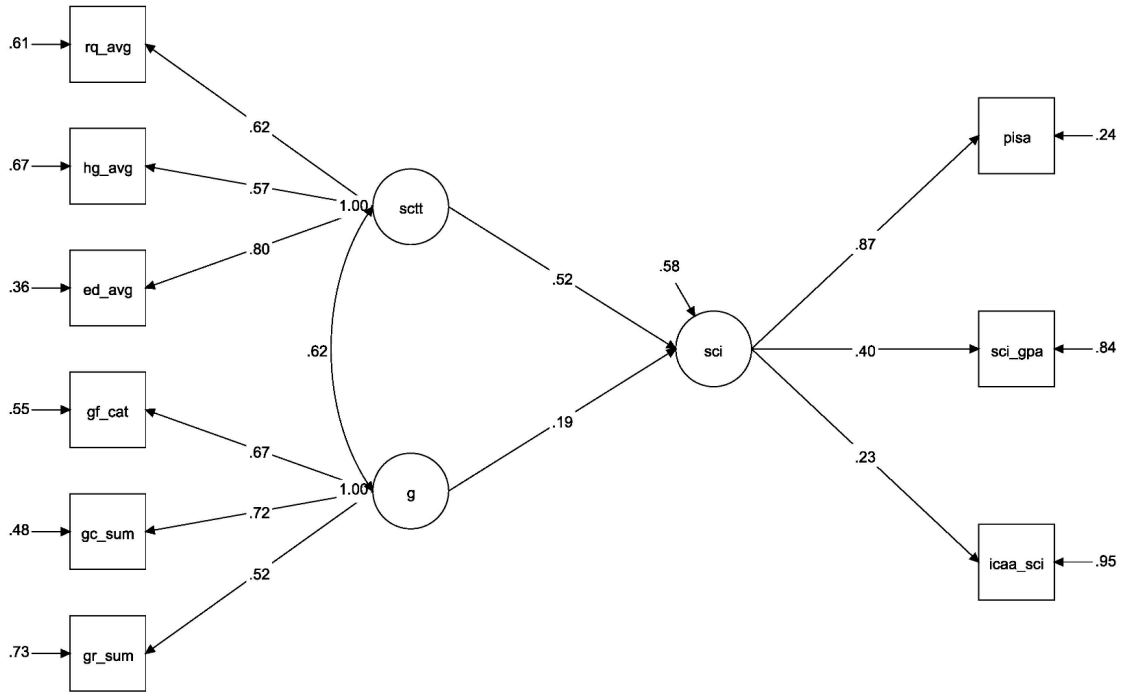


Note. rq_avg = research question, average rating; hg_avg = hypothesis generation, average rating; ed_avg = experiment design, average rating; gf_catl = fluid intelligence, Cattell's culture fair test score; gc_sum = crystallized intelligence, vocabulary advanced and extended range tests, summed; aut = alternate uses task, maximum associative distance; verb = verb generation, semantic distance; story = story task, divergent semantic integration; pisa = PISA total score; sci_gpa = science grades; icaa_sci = Inventory of Creative Activities and Achievements, science score.

We began by assessing how SCTT creativity scores relate to science achievement, verbal creativity, and general intelligence. A CFA with these four factors fit the data well: $\chi^2(48) = 71.73$, $p = .01$; CFI = .94; RMSEA = .05, 90% CI: [.02, .07]; SRMR = .05. All variables loaded onto their respective factors significantly (see Figure 5). Moreover, all latent variables were strongly correlated. Regarding science achievement, SCTT showed the highest latent correlation (.63), followed by general intelligence (.51) and verbal creativity (.50). Notably, general intelligence

Figure 6

Structural regression model from Study 2



Note. rq_avg = research question, average rating; hg_avg = hypothesis generation, average rating; ed_avg = experiment design, average rating; gf_catl = fluid intelligence, Cattell's culture fair test score; gc_sum = crystallized intelligence, vocabulary advanced and extended range tests, summed; pisa = PISA total score; sci_gpa = science grades; icaa_sci = Inventory of Creative Activities and Achievements, science score.

correlated highly with both SCTT (.63) and verbal creativity (.73), and interestingly, SCTT and verbal creativity were very highly correlated (.86). This suggests that, although SCTT correlated more strongly with science achievement than verbal creativity, SCTT and verbal creativity shared substantial variance, indicating domain-general contributions to the SCTT.

We then specified a regression model to test whether SCTT predicts science achievement controlling for intelligence: $\chi^2(24) = 39.46$, $p = .02$; CFI = .95; RMSEA = .06, 90% CI: [.02, .09]; SRMR = .05 (see Figure 6). With $R^2 = .42$, this model accounted for 42% of variance in science achievement. Critically, we found a large effect of SCTT on science achievement ($\beta = .52$, $p < .001$), and a small, nonsignificant effect of general intelligence ($\beta = .19$, $p = .23$). This was not the case, however, for verbal creativity: a separate regression analysis, with verbal creativity and general intelligence predicting STEM achievement (overall model $R^2 = .32$), $\chi^2(24) = 46.45$, $p = .004$; CFI = .89; RMSEA = .07, 90% CI: [.04, .10]; SRMR = .05, did not yield significant effects of either general intelligence ($\beta = .32$, $p = .27$) or verbal creativity ($\beta = .29$, $p = .38$), which were highly correlated ($r = .72$). Thus, a model incorporating the SCTT, as opposed to verbal creativity, accounted for an additional 10% of the variance in science achievement.

Finally, we evaluated measurement invariance across self-reported gender and found that the observed scores, assessed via Bayesian structural equation modeling, adhere to strict measurement invariance (please refer to the supplemental material file in the OSF for detailed findings).

Study 3

Study 2 provided evidence for the criterion validity of the SCTT, particularly its prediction of STEM achievement. SCTT performance was strongly correlated with verbal creativity and intelligence, yet it robustly predicted STEM achievement in a regression model controlling for intelligence. This was not the case for verbal creativity, which did not uniquely predict STEM achievement above intelligence.

In Study 3, we assessed the test-retest reliability of the SCTT. We used alternate test versions and administered them to undergraduate students one month apart. We expected to find a moderate to high test-retest correlation of human-rated creativity scores, based on the few studies that have assessed test-retest reliability of subjective creativity ratings on other creativity tests (Cropley, 2000). Finally, measurement invariance across gender was again explored.

Method

Participants

The total sample consisted of 181 undergraduates from Penn State University, who received optional course credit for their participation (mean age = 19.02, SD = 2.13; 143 females, 36 males, 1 other gender, 1 prefer not to disclose). They provided informed consent, and the study was approved by the PSU IRB.

Procedure

Participants completed two alternate versions of the SCTT about one month apart. The two versions consisted of two items per subtask (experiments, hypotheses, and questions); items were randomly selected to be on the two alternate tests. Participants were randomly assigned to complete either version first or second.

SCTT responses were scored for creative quality using the same subjective scoring method as before, with seven raters, using a planned missing data design. All measures were administered online via Pavlovia. Additional measures were collected and not analyzed here.

Results

Overall, participants produced 6,371 responses to the 12 SCTT items (6 items per participant, per form). The ICC was computed between the 7 raters across all items and subtasks. We found moderate reliability, ICC = 0.58, 95% CI: [0.56, 0.60], for the absolute agreement using a two-way model.

We began by examining the zero-order retest correlations at the subtask level. We found retest correlations ranging between .28 (for hypotheses) and .41 (for research questions). Further, the retest reliability of the observed overall scores was at .51, which was notably consistent across both versions of the SCTT (i.e.,as operationally defined by which of the alternate forms has been administered first).

Subsequently, we assessed time stability based on latent correlations in a CFA model where experiments, hypotheses, and research questions (average scores) were modeled as indicators for the higher-order factor. For timepoint 1, the model-based alpha reliability was .65, and omega reliability was .66. In comparison, for timepoint 2, the alpha reliability was .63 and omega reliability was .64.

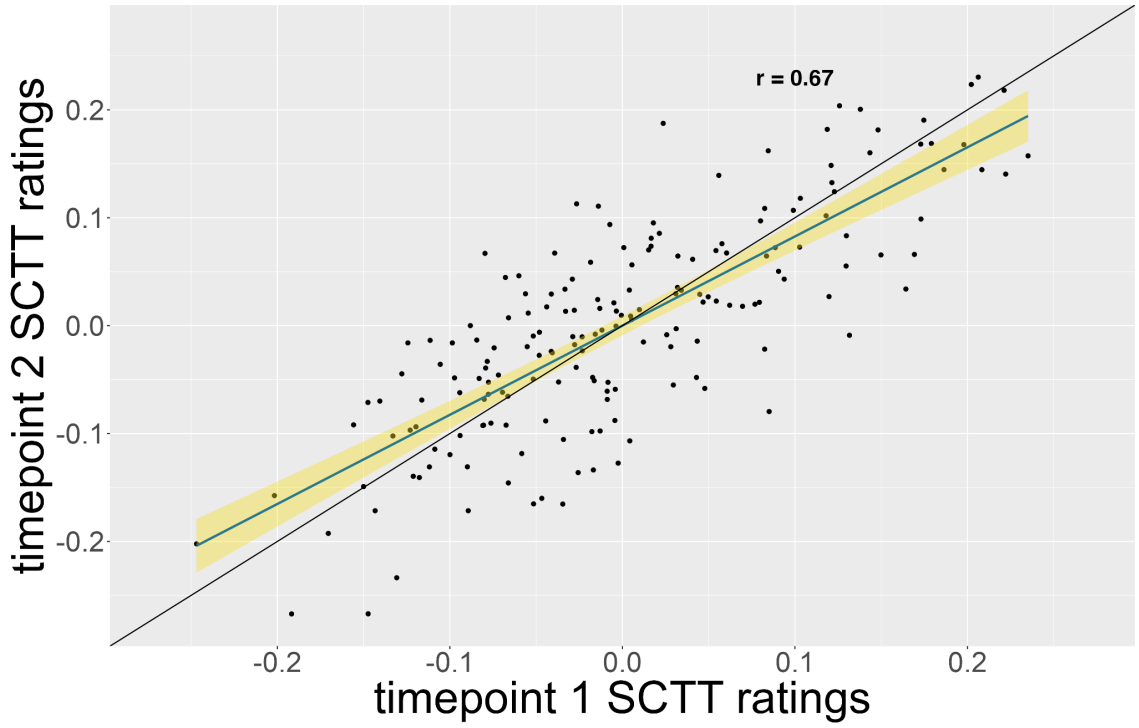
In addition, the empirical reliability for the factor scores was .69 at both timepoints. We also calculated the Factor Determinacy Index (FDI), which surpassed .80. This FDI score implies that the reliability was deemed sufficient for research purposes (Rodriguez et al., 2016).

Next, we assessed the test-retest reliability of the construct as indicated by the latent variable correlation within the model. The estimated latent variable correlation was found to be .67, which suggests a moderate level of stability over time (see Figure 6).

Lastly, measurement invariance analyses suggested that for the data in Study 3 strong invariance across gender was reasonable, yet not strict invariance. Thus, invariance was slgihtly lower as compared to Study 2 (please consult the supplemental material file in the OSF for detailed findings).

Figure 6

Test-Retest Correlation of SCTT Creativity Ratings



Note. *N* = 181. Black diagonal line reflects a perfect correlation between timepoints (*r* = 1.00).

Study 4

Study 3 evaluated the test-retest reliability of the SCTT. Alternate versions of the SCTT were constructed and administered to undergraduate students with a one month interval between sessions. We found moderate levels of stability, consistent with past work on subjective scoring methods (Barbot, 2019; Cropley, 2000; Long & Wang, 2022). The factor determinacy index indicated satisfactory reliability for research purposes at each measurement point.

Study 4 reexamined the question of measurement invariance of the SCTT at the item level. Measurement invariance was evaluated across gender groups for Study 2 and 3, with promising results. However, it was not possible to test invariance at the item-level for the missing data designs used in these studies, leaving it unclear if any demographic biases may be present. Furthermore, some gender groups were rather small and the groups were unbalanced. In Study 4, we also tested measurement invariance across different races and

ethnicities (Black, Hispanic, and White). Finally, to assess the SCTT's face validity from the test-taker's perspective, we administered a post-SCTT survey with questions regarding the degree to which participants viewed the SCTT as measuring their ability to think creatively in science.

Method

Participants

Participants were recruited online via Prolific to ensure demographic diversity in the sample. The following inclusion criteria were employed for all participants: student status (yes), fluent languages (English), and age (18-40). To assess language history, participants indicated whether English was their first language. We recruited six groups, combining two demographic variables: Race/ethnicity (Black, Latino/Hispanic, and White) and gender(female and male). The total sample consisted of 306 participants. Four participants did not report gender and race/ethnicity, so they were excluded from analyses. The final sample for analyses included 302 participants. The study was approved by the PSU IRB, and participants provided informed consent.

Procedure

Participants completed the SCTT, then answered survey questions related to demographics (race/ethnicity, and language history). All measures were administered online via Qualtrics.

SCTT. Participants completed the three SCTT subtests: research questions, hypotheses, and experiments. We included three items from each subtest, selected based on factor loadings from Study 1; all participants completed the same items (see OSF for items). SCTT subtests were presented in a random order, with items presented in a fixed order within subtests. As in Study 1, for each item, participants produced three responses. These responses were automatically scored for creative quality using the RoBERTa scoring model (see Supplemental Method; Appendix B).

Face Validity. Participants completed three items to assess SCTT's face validity using a 1 (*strongly disagree*) to 5 (*strongly agree*) scale: *1) Coming up with hypotheses, research questions, and experiments in this test reflects important skills for being creative in science. 2) The kinds of thinking needed for this test appear related to doing creative scientific research. 3) I felt like this test allowed me to show my ability to think creatively about science.*

Measurement Invariance Testing

We employed the criteria for measurement invariance testing proposed by Chen (2007). For sample sizes smaller or equal to 300, Chen proposed to consider decreases in the CFI by more than .005 to be indicative of a violation of measurement invariance. We further checked increases in RMSEA by more than .010 and increases in the SRMR by more than .025 for evaluation of weak invariance, as well as increases in RMSEA by more than .010 and increases in the SRMR by more than .005 for evaluation of strong and strict invariance.

Results

First, we assessed the face validity of the SCTT from the test-taker's perspective. We observed strong face validity on the post-SCTT survey questions. All means (all *M*s > 4.02) of the three items were significantly larger than the scale midpoint of 3 (all *p*s < .001; see Table 2).

Table 2

Descriptive Statistics for Face Validity Questions

Question	Mean (sd)	Min	Max
Coming up with hypotheses, research questions, and experiments in this test reflects important skills for being creative in science.	4.38 (.70)	1	5
The kinds of thinking needed for this test appear related to doing creative scientific research.	4.17 (.84)	1	5
I felt like this test allowed me to show my ability to think creatively about science.	4.02 (.92)	1	5

Measurement Invariance across Gender

Regarding measurement invariance, we first evaluated invariance across gender groups. The configural three-dimensional model fit well in both groups (women: CFI = .952, RMSEA = .064, SRMR = .045; men: CFI = 1.00, RMSEA = .000, SRMR = .031). Table 3 displays the excellent fit of the configural model as a multi-group model. Comparing this model with a weak invariance model did not violate the primary change in CFI criterion (CFI dropped only by .003). Next, the comparison between the weak invariance model and the strong invariance model also did not show any violations of measurement invariance (CFI dropped by .002). Finally, strict measurement invariance could also be assumed across gender groups as CFI even increased compared to the strong invariance model. Notably, the supplemental criteria for RMSEA and SRMR also fulfilled the measurement invariance criteria. Thus, measurement invariance of loadings, intercepts, and residual variances could be assumed across sex groups.

The achieved level of invariance allowed exploring latent mean differences. We did not observe any significant mean differences between the sex groups on any of the three latent variables.

.Table 3

Measurement Invariance Findings across Gender Groups

Model	**CFI**	**RMSEA**	**SRMR**
Configural	.990	.030	.038
Weak	.987	.033	.048
Strong	.985	.033	.051
Strict	.995	.019	.051

Measurement Invariance across Race/Ethnic Groups

Next, we evaluated measurement invariance across ethnic groups. The configural three-dimensional model fitted well in all race/ethnic groups (Black: CFI = .993, RMSEA = .027, SRMR = .050; Hispanic/Latino: CFI = .962, RMSEA = .053, SRMR = .050; White: CFI = 1.00, RMSEA = .000, SRMR = .042). Table 3 shows the excellent fit of the configural model as a multi-group model. Comparing this model with a weak invariance model did not violate the primary change in CFI criterion (CFI dropped only by .003). Next, the comparison between the weak invariance model and the strong invariance model did not show any violations of measurement invariance (CFI even increased). Finally, strict measurement invariance could also be assumed as CFI increased again as compared to the strong invariance model. Notably, the supplemental criterion for RMSEA was also fulfilled. Thus, measurement invariance of loadings, intercepts, and residual variances was also reasonable across ethnic groups.

We also assessed latent mean differences in SCTT scores across ethnic groups. We found that Hispanic participants performed significantly better compared to Black participants on experimental design (latent mean difference = 0.034, z = 2.75, p = .006). In addition, Whites performed significantly better on experimental design as compared to Black participants (latent mean difference = 0.039, z = 3.04, p = .002). These differences can likely be explained by differences between the race/ethnic groups in terms of reporting English as a first language. The distribution of this variable differed significantly between the ethnic groups ($\chi^2(2) = 97.14$, $p < .001$), with Black participants reporting more often that English was not their first language.

Table 3

Measurement Invariance Findings across Ethnic Groups

Model	**CFI**	**RMSEA**	**SRMR**
Configural	.988	.033	.047
Weak	.985	.034	.067

Strong	.994	.021	.069
Strict	1.00	.000	.073

Study 5

Study 4 addressed test fairness across race/ethnicity and gender groups, providing evidence for strict measurement invariance and strong face validity. These findings extend our earlier studies indicating that the SCTT is a reliable and valid measure of scientific creative thinking that functions similarly across diverse groups of participants.

In Study 5, a method for automatically scoring SCTT responses was tested using Llama-2, an open-source language model. Llama-2 is a large pre-trained transformer model from Meta AI. A specialized version, Llama-2-Chat, underwent additional training by Meta AI on human feedback for dialogue alignment. Both models demonstrate strong performance on benchmarks compared to models like ChatGPT (Touvron, Martin, et al., 2023).

We fine-tuned the base Llama-2 model and Llama-2-Chat using SCTT responses and human creativity ratings from Studies 1-3 (*N* = 18,660 responses). Our primary objectives were to evaluate model prediction of new creativity ratings, and importantly, generalization to new SCTT items, allowing future items to be developed and automatically scored. Llama-2-Chat's additional human feedback training was expected to potentially yield improved alignment with human SCTT ratings. Further, each model was compared to a word count baseline to evaluate if the models's predictive capacities extended beyond elaboration alone.

Method

Automated Scoring Models

Llama-2-7B is a 7 billion parameter model pre-trained by Meta AI. Llama-2-7B-chat is identical to Llama-2-7B except it underwent additional supervised training by Meta AI to adapt it into a chatbot that produces safe and useful responses. Although the smallest in the Llama-2 family, both 7B variants have demonstrated strong performance on natural language

processing benchmarks (Touvron, Lavril, et al., 2023). To enable prediction of continuous creativity ratings, a regression layer was added to the output of each model to generate a single scalar rating for input for each SCTT response.

Data Preprocessing

An initial set of 18,660 creativity-rated responses were collected from Studies 1-3 and an earlier pilot study. Ratings were transformed based on Judge Response Theory (JRT; Myszkowski & Storme, 2019) θ (factor) scores using a Graded Response Model (Samejima, 1997) and normalized to a 0-1 range. JRT is an extension of Item Response Theory to rating data that taps the underlying construct of creativity and helps adjust for issues like rater severity. Using an 'exact match' deduplication strategy, identical responses were consolidated into single entries with averaged θ scores (Organisciak et al., 2022).

The resulting 18,161 responses were divided into training (70%), validation (10%), and held-out (20%) sets. The training set provides responses for the model to learn; the validation set determines optimal model settings; the held-out test set evaluates best model generalization. In addition, a prompt transfer subset (Zhou, 2021) using the prompts "food" and "building" (1,062 responses) was used to assess generalization to new prompts the model was never trained on.

Automatic Scoring Procedure

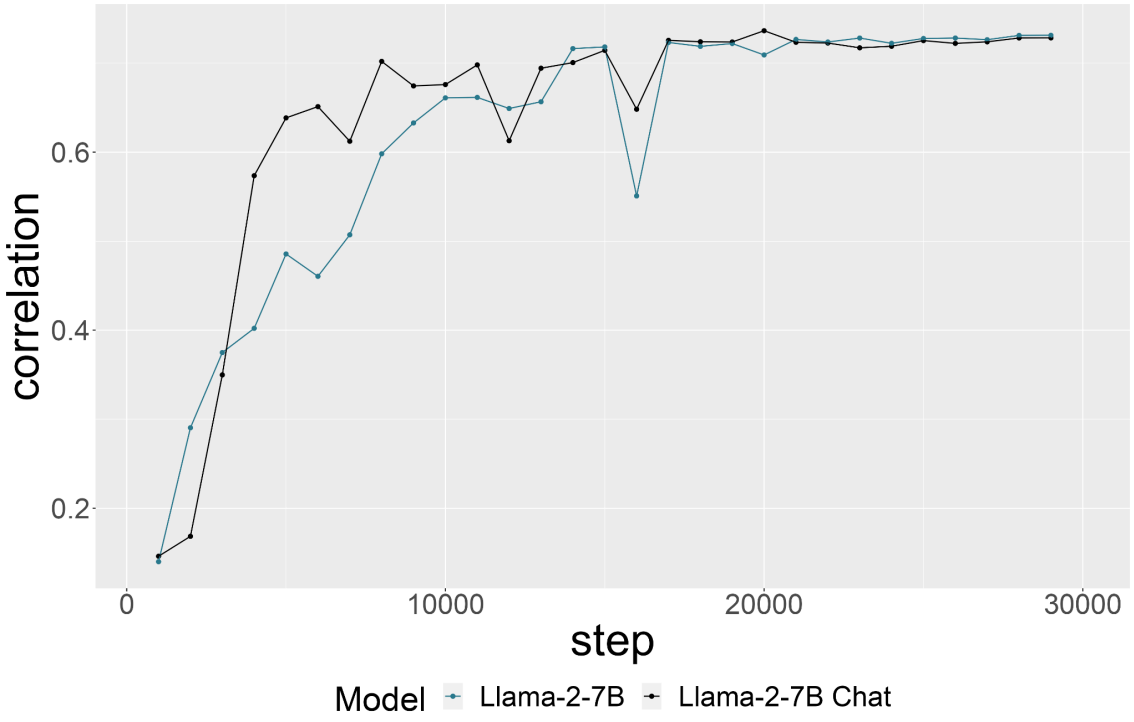
Given the computational demands of fine-tuning 7B parameter models, Low Rank Adaptation (LoRA; Hu et al., 2021) was employed rather than full fine-tuning. LoRA freezes all model weights and injects a small number of trainable matrices that influence output; this enables efficient large model training, with performance comparable to full fine-tuning (Hu et al., 2021). During training, LoRA matrices were updated to minimize the error between predicted and human (JRT) creativity ratings. We used a LoRA r factor of 4, alpha of 32, and 10% dropout probability for regularization (see Hu et al., 2021). Models were trained for 10 epochs with a learning rate of $5e^{-5}$ and a batch size of 4. Checkpoints were saved every 1,000

steps (4,000 responses, given the batch size of 4), with the optimal checkpoint selected by maximizing the human-model Pearson correlation in the validation set.

Results

Figure 7

Model-Human Validation Correlation Across Training Steps



Note: Llama-2-7B is reflected by the green line while Llama-2-7B-chat is reflected in black.

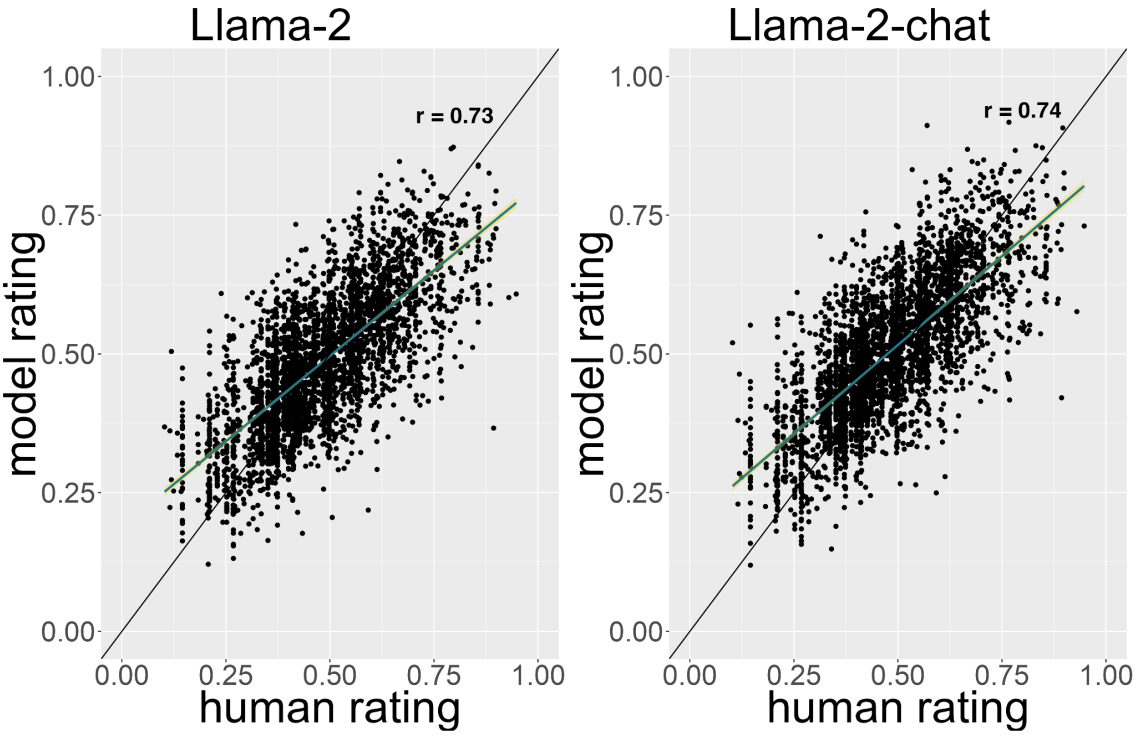
Pearson correlation on the validation set across training can be seen in Figure 7. The training time course showed that Llama-2-Chat learned to predict SCTT creativity ratings faster than base Llama-2. The human-model correlation was maximized on the validation set at the 29,000th training step for Llama-2 and the 20,000th training step for Llama-2-Chat. However, both models achieved similar performance in predicting human creativity ratings.

Applying the best-fitting checkpoint of each model to the held-out test set, we found strong correlations between model-predicted and human creativity ratings for both Llama-2 (r = .73; 95% CI: [.72, .75]) and Llama-2-Chat (r = .74; 95% CI: [.72, .75]), as seen in Figure 8.

Critically, each model substantially exceeded the elaboration baseline based on the correlation between word count and human ratings (r = .30; 95% CI: [.27, .33]), indicating each learned a solution more sophisticated than a simple 'more is better' rule.

Figure 8

Held-out Test: Model Predictive Accuracy By Rating Value



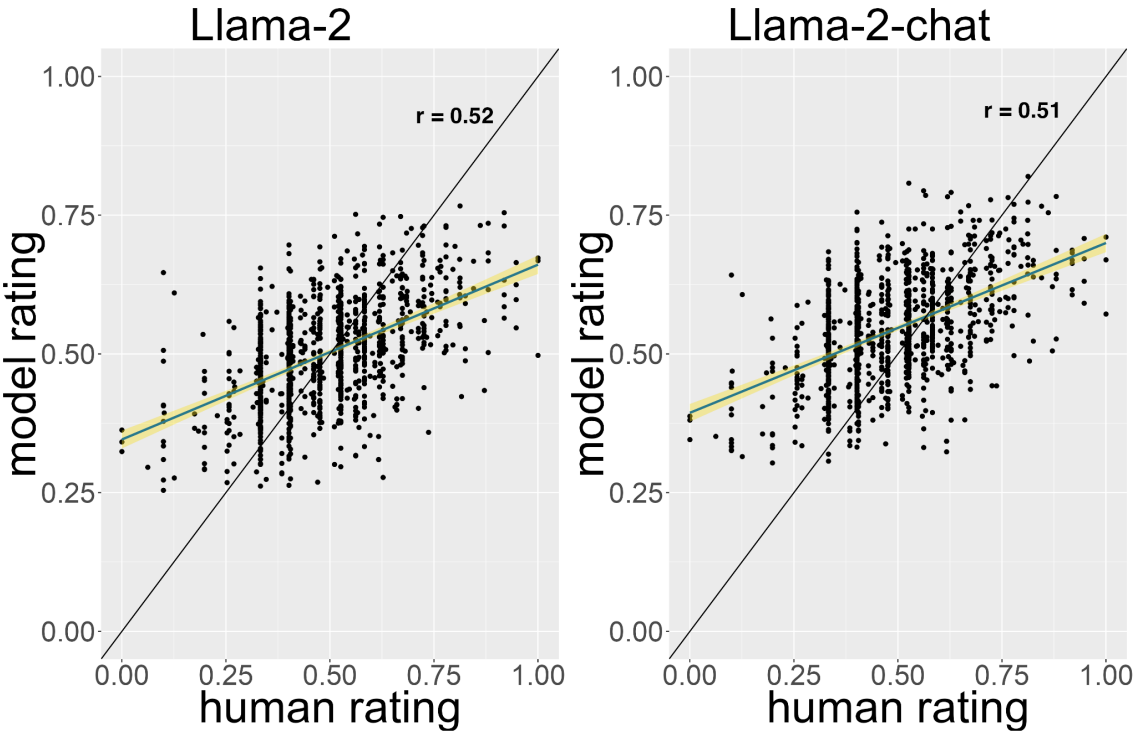
Note: The green line in each sub-plot reflects the line of best fit with 95% CI in yellow shading. The diagonal black line reflects perfect performance.

We then tested each model's ability to accurately generalize to responses from untrained prompts, using the prompt transfer dataset. Although the models were not trained on the "food" and "building" prompts, each exhibited strong generalization (Figure 9): Creativity ratings from Llama-7B (r = .52; 95% CI: [.47, .56]) and 7B-chat variant (r = .51; 95% CI: [.47, .56]) closely aligned with human ratings. Notably, elaboration/word count did not significantly correlate with human ratings (r = .02; 95% CI: [-.04, .08]). Thus, the models

successfully generalized to untrained prompts—without relying on simple elaboration—suggesting they acquired fine-grained representations of how humans rate creativity on the SCTT.

Figure 9

Prompt Transfer: Model Predictive Accuracy By Rating Value



Note: The green line in each sub-plot reflects the line of best fit with 95% CI in yellow shading. The diagonal black line reflects perfect performance.

Discussion

Fostering creativity in STEM education requires reliable and valid tools to assess scientific creativity. In the present research, we developed and evaluated the Scientific Creative Thinking Test (SCTT) in five studies. Study 1 assessed the internal reliability of the SCTT, finding that items loaded significantly onto hypothesized factors (research questions, hypotheses, and experiment designs). Study 2 provided evidence for criterion validity, with SCTT creativity scores predicting science achievement beyond intelligence. Study 3 found moderate 1-month

test-retest reliability for the SCTT using alternate forms. Study 4 demonstrated measurement invariance across gender and race/ethnicity, as well as face validity from the student perspective. Study 5 validated an automated scoring system for the SCTT using fine-tuned language models. The current work thus demonstrates that the SCTT can measure the scientific creative potential of young adults in a reliable, valid, and efficient way.

The SCTT joins existing tests that assess scientific creativity, including the Scientific Creativity Test (SCT; Hu & Adey, 2002), the Creative Scientific Ability Test (C-SAT; Sak & Ayas, 2013), and the Divergent Problem Solving Ability in Science Test (DPAS; Aschauer et al., 2022). Although all tests assess scientific creativity through divergent thinking tasks, there are key differences in their target populations and scoring approaches. Regarding populations, to our knowledge, the SCTT is the first to be thoroughly validated for use in undergraduate students (cf. Sternberg et al., 2020 for a psychology-specific creativity test). Regarding scoring, other tests tend to focus on fluency and flexibility, whereas the SCTT focuses exclusively on creative quality, due to psychometric issues associated with fluency-based scoring (Forthmann et al., 2020). Our decision to focus on creative quality was also based on previous validation work on divergent thinking, showing stronger validity evidence for creativity ratings over fluency and flexibility, including predicting real-world creative achievement (Plucker et al., 2011). The SCTT therefore complements existing scientific creativity tests, with evidence for its reliability and validity across three studies.

Study 1 aimed to evaluate the reliability of the SCTT. We tested five items per competency—research questions, hypotheses, and experiments—and assessed how well they loaded onto their respective factor, as well as a higher-order factor. Results showed the 15 SCTT items loaded significantly on their respective competency factors and, when modeled together, the three factors loaded strongly on a higher-order general SCTT factor. The general factor aligns with prior research on the shared variance across different types of creativity assessments (Ayas & Sak, 2014).

The SCTT's reliability was further assessed in Study 3, where we evaluated test-retest reliability using alternate test versions administered about a month apart. Test-retest reliability was found to be adequate (classical retest reliability was r = .51), with a latent variable correlation of .67 (reflecting time stability of the construct) between administrations and an empirical reliability of .69 for the factor scores, which surpassed the threshold of .80 on the Factor Determinacy Index (Ferrando & Lorenzo-Seva, 2018). This finding aligns with a recent meta-analysis of test-retest studies using subjective creativity scoring, which found a mean of .61 across tests (Long & Wang, 2022). The time stability implies a certain degree of malleability in scientific creative thinking ability, which is encouraging for efforts to foster creative thinking through instruction and interventions. While perfect construct stability over time would not be ideal, the evidence for moderate stability suggests scientific creative thinking is a skill that can be developed, rather than a fixed trait.

Importantly, our measurement invariance findings across gender groups indicated that the psychometric properties of the SCTT were rather constant, which is an essential precondition for fair measurement across demographic groups. We also assessed measurement invariance across race/ethnic groups in Study 4. Overall, we did observe any strong hints that the SCTT could potentially be biased by varying psychometric properties (and therefore, substantive interpretation of the underlying constructs) across demographic groups. Notably, the alternate form design of Study 3 did not permit testing measurement invariance of the SCTT across timepoints. Given the use of different items at each timepoint, establishing invariance is not feasible. This precluded examining potential differential item functioning over time. An important direction for future research will be developing additional alternate forms of the SCTT with overlapping items, which would allow invariance testing and provide further evidence for the stability of measurements. For now, our alternate forms design provides initial evidence for test-retest reliability, complementing the internal consistency assessed via factor analysis in Study 1. But fully demonstrating measurement invariance across multiple

timepoints remains an important goal for further establishing the psychometric strength of the SCTT, despites numerous known challenges to establish measurement equivalence (Barbot, 2019).

Study 2 demonstrated the validity of the SCTT for predicting STEM achievement. Importantly, SCTT creativity ratings explained significant variance in STEM achievement beyond general intelligence. This was not the case for verbal creativity, however, which did not predict STEM achievement in a regression with intelligence—despite verbal creativity being highly correlated with SCTT performance. This divergence suggests the SCTT taps domain-specific creative thinking skills in science over and above general verbal creativity. In contrast, generating scientific hypotheses and experiments may involve additional domain-specific knowledge and cognitive processes, such as conceptual knowledge of scientific phenomena. These results highlight the importance of using domain-specific assessment to measure creative thinking in STEM.

The large correlations of both SCTT and verbal creativity with general intelligence is consistent with past work on intelligence and creativity (Silvia, 2015). Studies of domain general creative thinking have found that intelligence relates to creative thinking through executive control (Benedek et al., 2014; Frith et al., 2021), allowing people to strategically search through memory, inhibiting obvious/unoriginal ideas along the way. Notably, executive control has also been linked to domain-specific creative writing (Johnson et al., 2022; Taylor & Barbot, 2021). In the context of scientific creativity, intelligence may support controlled generation and evaluation of ideas via strategic memory search and inhibition over domain-specific knowledge, consistent with recent fMRI work suggesting interactions of executive control and memory retrieval during hypothesis generation (Beaty et al., 2023).

Regarding automated scoring of the SCTT, Study 4 found that fine-tuning both the base Llama-2 and Llama-2-Chat variants yielded strong predictions of human creativity ratings. Interestingly, the chat variant showed faster learning of human creativity ratings—perhaps due

to its additional human feedback training—yet the two models ultimately produced nearly identical predictions of human ratings in the held-out set. The prediction magnitude far exceeded an elaboration baseline, indicating the models did not rely on a simple "more is better" rule and that they had learned more sophisticated representations of how humans evaluate the creative quality of scientific ideas. Importantly, both models demonstrated generalization to new SCTT prompts that were not present in their training set. In the future, additional items could be developed and automatically scored by these models. However, they should be carefully constructed to match the form and content of the SCTT, and ideally, undergo psychometric evaluation to ensure reliability and validity.

To our knowledge, study 4 is the first demonstration of automated assessment of domain-specific creativity tasks using large language models. Our work extends recent approaches to score the domain-general alternate uses task using GPT-3 (Organisciak et al., 2022), which is a very large proprietary model from Open AI. In contrast, we show that Llama 7B—an open-source model orders of magnitude smaller than GPT3—can be fine-tuned to automate scoring for domain-specific creativity tests.

Limitations and Future Directions

Our study has a few limitations worth noting. First, the SCTT would benefit from additional psychometric analysis of content/face validity. We encourage researchers interested in specific STEM fields (e.g., chemistry, biology) to develop content-specific test items in consultation with teachers to ensure content/face validity. The SCTT is intended to measure general thinking skills for scientific creativity across multiple domains; for pedagogical purposes, developing domain-specific test items would be worthwhile. For now, Study 5 provides initial content validity evidence from the student perspective. Students from diverse backgrounds agreed that the SCTT was measuring their own abilities to think creatively in a scientific context.

Regarding human rating of SCTT responses, it is important to note the rating scale and instructions provided in Study 1, while intended to capture originality as the key component of divergent thinking, may have still allowed subjective assessment criteria related to plausibility. Further efforts are needed to better direct raters to judge relevance, effectiveness, and appropriateness when evaluating responses in order to ensure ratings capture both originality and effectiveness, rather than just uniqueness. This could be achieved by rating ideas separately for originality and effectiveness, with automated scoring methods employed to evaluate both dimensions (cf. Luchini et al., 2023).

Several opportunities exist to extend the validation and application of the SCTT in future work. One possible direction is adapting the SCTT for specific STEM domains, such as physics, chemistry, and engineering. Domain-specific versions could better align with undergraduate majors and help evaluate creative growth within degree programs. The SCTT also provides an outcome measure for testing pedagogical interventions aimed at enhancing scientific creativity, such as Flex-based learning (Haim & Aschauer, 2022). Another promising direction is incorporating scientific creativity into undergraduate and graduate admissions. With the SAT and GRE being discontinued in US universities, creativity tests present a more equitable alternative that tends to be less biased by race/ethnicity and socioeconomic background (Kaufman, 2010).

To enable widespread adoption, continued optimization of automated scoring will be important going forward. We provide open access to the SCTT training data and fine-tuned language models to support this goal. Researchers can determine if smaller, open-source models like BERT (Devlin et al., 2019) perform comparably to large proprietary models such as GPT-4, potentially allowing them to be run for free on personal computers. Future work should also explore whether curating the training set of creative responses, for example, by including only highly reliable creativity ratings, can improve the ability of language models to generate reliable creativity metrics, as has been shown in other domains (Gunasekar et al., 2023). We

look forward to continued research refining the SCTT and other STEM-based creativity tests to maximize their impact for STEM education.

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Appendix A

SCTT Items from Study 1

Items followed by 'SELECTED' in brackets were chosen for the final item set.

Experiments

Your friend thinks only water and sunlight determine how much a flower grows, and nothing else matters. How could you test that hypothesis?

Your cousin claims they can read minds, but it only works when a person doesn't know their mind is being read. How could you test to see if this super power is real? **[SELECTED]**

You think students at your school are friendlier than students at most schools. How could you test that hypothesis? **[SELECTED]**

You think some animals have a sense of humor that humans don't usually understand. How could you test that hypothesis? **[SELECTED]**

You think birds can understand humans but just choose to ignore us because what humans say isn't important to their lives. How could you test that hypothesis? **[SELECTED]**

Hypotheses

You just started going to a new school. You notice that more students sing in the hallways at this school than at your old school. What hypotheses do you have about why that is? **[SELECTED]**

You notice that dogs seem to like one of your friends, but cats seem to like another friend. What hypotheses do you have about why that is? **[SELECTED]**

THE SCIENTIFIC CREATIVE THINKING TEST (SCTT)

You notice that the water in one lake is warmer than the water in another lake even though they both get the same amount of sunlight. What hypotheses do you have about why that is? **[SELECTED]**

On a field trip, you drive past a massive field with hundreds of large holes visible as far as the eye can see. What hypotheses do you have about what purpose the holes may serve?

You look outside one night and see stars disappearing one by one from the sky. What hypotheses do you have about why that is? **[SELECTED]**

Research Questions

You are introduced to a robot that can learn and think like humans. What scientific questions could you ask about this?

You travel on a spaceship to a new planet outside of our galaxy. What scientific questions could you ask about this planet? **[SELECTED]**

You created the first vehicle that can reach an unexplored part of the ocean floor. What scientific questions could you ask about this part of the ocean floor? **[SELECTED]**

You travel to a jungle that contains no human life and is completely unknown to the scientific community. What scientific questions could you ask about this jungle? **[SELECTED]**

You discover a remote island with people who communicate without speaking. What scientific questions could you ask about this? **[SELECTED]**

Appendix B

Supplemental Method and Results for RoBERTa SCTT Scoring Model

Study 5 demonstrated the ability of Llama-2 to accurately match human creativity ratings for responses to SCTT items—both items the model was explicitly trained on as well as items that were omitted from the training data. While performant, Llama-2 7B is a resource-intensive model to train and use. Due to its 7-billion-parameter size, roughly 30GB of RAM is required to run the model and several multiples of that figure to train the model. These hardware requirements are outside the range of compute many researchers are likely to have access to, which is undesirable from impact and open-science perspectives alike.

Consequently, we investigated whether a fine-tuned version of RoBERTa-base (Liu et al., 2019) could achieve comparable performance to Llama-2. RoBERTa-base is only 125 million parameters and assumes a memory footprint of only ~500MB. As such, the model is compatible with all contemporary consumer computers, making it more accessible to other researchers and educators.

Method

The method was largely the same as in Study 5. We used the exact same preprocessed training, validation, test, and prompt transfer datasets used in Study 5. However, given the lower computational burden of training RoBERTa-base, we performed a hyperparameter search to find optimal settings and to give the RoBERTamodel the best chance of matching the performance of Llama-2. We searched over the following hyperparameters: learning rate (1e-7 to 1e-3), training batch size (2, 4, 8, or 16), and training set augmentation (base, expanded, filtered_base, or filtered_expanded). Training set augmentation was an experimental manipulation with four levels. The ‘base’ training dataset was identical to that of Study 5. The

'expanded' dataset was a version of the base dataset, where each item appeared four times: once with the base dataset's sentence frame adjective (i.e., *A creative* hypothesis for why there are massive holes in a field is RESPONSE) and three more times, each with a different adjective (*surprising, intriguing,* or *interesting*). The 'filtered_base' dataset sought to remove responses from the base training dataset that creativity raters disagreed about; we set a fairly strict threshold for inclusion and included only responses that had a mean pairwise rater disagreement of less than 1. The 'filtered_expanded' dataset expanded the filtered dataset in the same way as the 'expanded' dataset.

For the hyperparameter search, we fixed the number of training epochs at 40 and allowed the Tree-Structured Parzen Estimator (TPE) to find hyperparameter values that minimized MSE over that training timeframe. TPE is a Bayesian method for hyperparameter search that learns from previously-tested hyperparameter combinations and proposes new combinations that are likely to improve the target metric (in our case, MSE). We set the number of trials (i.e., TPE-selected hyperparameter combination) to 250. Following the hyperparameter search, we trained the best-fitting model and saved checkpoints every 1000 training steps. We then looped over those checkpoints to determine which maximized Pearson correlation between model scores and JRT theta scores on the validation set.

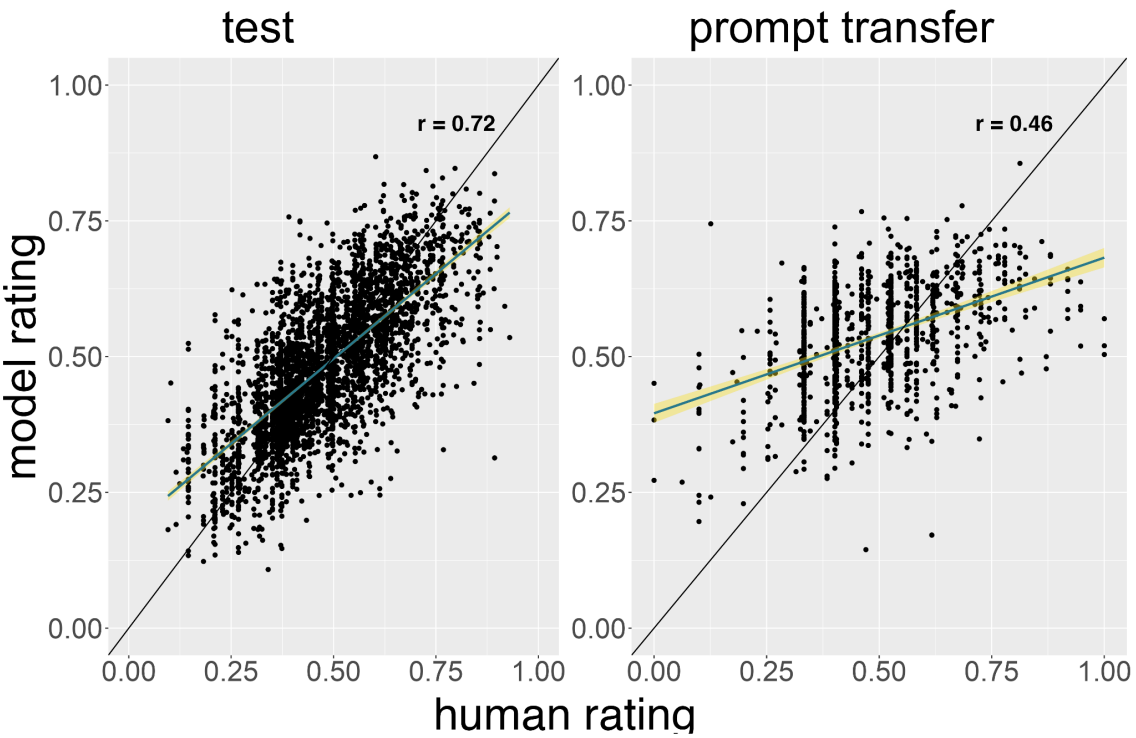
Results

The hyperparameter search found the best model achieved a tight fit to the validation set (MSE = .0101). The TPE found that a learning rate of 5.72e-6, a batch size of 16, and the 'expanded' dataset led to the best human-model fits on the validation set. The follow-up training duration search found that the Pearson correlation between human and model-predicted scores was highest at the 62,000th training step (~20 epochs; r = .734).

Applying the optimized model to the held-out test dataset, we found the model's predictions correlated strongly with human creativity ratings (r = .72; 95% CI [.70, .73]; Figure B1)—far surpassing the correlation between word count and human creativity ratings (r = .30; 95% CI: [.27, .33]). We then assessed the model using the more challenging prompt transfer dataset; again, we found strong fits to human creativity ratings (r = .46; 95% CI [.41, .50]; Figure B1). As with the held-out test set, the model's fit to human ratings greatly exceeded that of word count to human ratings (r = .26; 95% CI [-.04, .51]).

Figure B1

Model-Human Fits for Held-out Test and Prompt Transfer Datasets



Summary

Here, we asked whether it was possible to develop a performant but resource-efficient tool for scoring SCTT responses—aiming to match the performance of Llama-2 as closely as possible but using a model that fits within researchers' computational budgets. Surprisingly, we found that the fine-tuned RoBERTa model was competitive with Llama-2, despite being 2% the size. This was especially true for the test set, on which RoBERTa was only ~1% worse than Llama-2. RoBERTa also performed quite well on the prompt transfer set, but at a larger discount relative to Llama-2 (~11% worse).

Overall, we found that RoBERTa is a suitable and computationally-cheap replacement for Llama-2 when scoring new responses to prompts the model was trained on. At only a 11% discount, RoBERTa may also be a viable stand-in for Llama-2 replacement for prompt generalization for those on a computational budget—though, the larger fine-tuned model (Llama-2) is preferable when the goal is to generalize to novel/untrained SCTT prompts.